

Nithish S B

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Profile Summary

Artificial Intelligence and Machine Learning graduate with hands-on experience building AI applications using Python, PyTorch, FastAPI, and modern LLM frameworks. Skilled in Retrieval-Augmented Generation (RAG), reinforcement learning, deep learning, and computer vision. Passionate about developing scalable AI systems for real-world applications.

Education

Sri Ramachandra Institute of Higher Education and Research

Oct 2022 - Aug 2026

B.Tech in C.S.E (Artificial Intelligence and Machine Learning) | CGPA: 8.50

Projects and Internship

AI Intern - AI Doctor Copilot using Retrieval-Augmented Generation (RAG)

Apr 2026 - Jun 2026

SRIHER(DU)

- Developed a clinical decision support system using Llama 3, Ollama, FAISS, and Retrieval-Augmented Generation (RAG) to provide evidence-grounded medical responses.
- Built a semantic retrieval pipeline using all-MiniLM-L6-v2 embeddings with top-k document retrieval from curated medical PDFs.
- Implemented a multi-layer safety framework including emergency symptom detection, confidence scoring, mandatory source citation, multilingual support (English, Hindi, Tamil), and medical disclaimers.
- Reduced factual hallucinations by approximately 80% compared to a baseline LLM through retrieval-based generation.

AI Intern - Adaptive Traffic Signal Control using Reinforcement Learning

Feb 2026 - Apr 2026

SRIHER(DU)

- Conducted a comparative study of Q-Learning, Deep Q-Network (DQN), and Double DQN for adaptive traffic signal optimization.
- Simulated a four-way traffic intersection using SUMO with a Python fallback environment for reproducible experimentation.
- Evaluated models using reward, queue length, waiting time, and signal phase transitions, with Double DQN achieving the best overall performance.
- Designed a modular RL framework supporting future extensions to multi-intersection traffic systems.

ML Intern - Automated Brain Tumor Segmentation from MRI Scans

May 2025 - Jun 2025

SRIHER(DU)

- Implemented an automated brain tumor segmentation system using a U-Net based CNN architecture.
- Segmented tumor regions including whole tumor, tumor core, and enhancing tumor.
- Evaluated model performance using Dice Coefficient and Intersection over Union (IoU) metrics.

ML Intern - Leukemia Detection Using CNN

Aug 2024 - Oct 2024

SRIHER(DU)

- Developed a deep learning-based leukemia detection system using Convolutional Neural Networks (CNN) with VGG19 architecture.
- Performed image preprocessing and data augmentation to improve dataset quality and model generalization.
- Trained and evaluated the model on microscopic blood smear images to classify various stages of Leukemia.

Technical Skills

Languages: Python, SQL, TypeScript

AI / Machine Learning: PyTorch, Scikit-learn, Hugging Face Transformers

LLMs & AI Frameworks: LangChain, LangGraph, Prompt Engineering, Retrieval-Augmented Generation (RAG), OpenAI API, Ollama

Backend & APIs: FastAPI, REST APIs, Uvicorn

Databases & Vector Stores: PostgreSQL, FAISS, Redis

Deep Learning & Computer Vision: CNNs, OpenCV

Tools: Git, GitHub, Docker, Jupyter Notebook, Postman

Cloud & Deployment: Vercel, Hugging Face Hub